

E-commerce Delivery & Customer Satisfaction — Insight Summary

Olist Public Dataset · 96,470 delivered orders · Brazil, 2016–2018

96,470 Total Delivered Orders	4.14 / 5 Avg. Review Score	6.77% Orders Delivered Late	health_beauty Top Category by Revenue
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Key Findings

- Late deliveries devastate satisfaction: orders delivered late average a 2.26/5 review vs. 4.28/5 for on-time orders — a 2-point gap. 62.9% of late orders receive a 1–2 star review, vs. only 9.5% of on-time orders.
- Delay is concentrated in time, not spread evenly: late-delivery rate spiked to 18.96% in March 2018, 14.13% in February 2018, and 12.40% in November 2017 (Black Friday period) — versus a low single-digit baseline most other months, pointing to seasonal/carrier capacity strain rather than a chronic system-wide problem.
- A small set of sellers drives disproportionate delay: the worst-performing seller (11+ orders) averages 9.6 days late with 36% of orders late, while the large majority of sellers ship close to on time — the problem is concentrated, not universal.
- Geography matters less than expected: state-level average review scores stay in a fairly narrow 3.8–4.2 band (MA, AL, PA lowest), suggesting delivery delay — not region — is the primary satisfaction driver.

Recommendations

1. Flag and intervene on chronically late sellers.

Build a monthly seller scorecard (avg. delay days, % late, order volume) and set an action threshold — e.g., >25% late orders with 10+ shipments — that triggers a support call, tighter SLA, or reduced marketplace visibility. This targets the sellers already identified as the worst offenders instead of treating delay as a uniform, marketplace-wide issue.

2. Add capacity buffers ahead of peak periods (Nov–Dec, and the post-peak Feb–Mar backlog).

Since the late-delivery rate roughly triples during and just after Black Friday/holiday season, renegotiate carrier capacity or extend estimated delivery windows proactively for that period, rather than letting the promised date slip after the fact — which is what most damages review scores.

3. Tighten estimated-delivery-date accuracy as a satisfaction lever, not just a logistics metric.

Because the review-score gap between late and on-time orders is so large (2 points), a more conservative/accurate estimated delivery date — even if slightly longer — likely improves satisfaction more than shaving actual transit time. Test this by A/B-widening estimates for high-risk seller/season combinations and tracking review-score lift.

Suggested Next Test

Cheapest validation: pull the worst-performing sellers identified in this analysis, apply a stricter estimated-delivery buffer (+3–5 days) to their listings only for one peak month, and compare their late-rate and review-score shift against a control group of similar sellers with no change.